

Hyo-jeong (Tiffany) Lee

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Summary

Computational neuroscientist working on **neural network simulations**, **neural manifolds**, and **cognitive modeling**, with a focus on understanding the cognitive interpretation of neural dynamics during learning.

Education

Gwangju Institute of Science and Technology (GIST) Mar 2022 – Dec 2026

B.S. in Electrical Engineering & Computer Science, Minor in Biomedical Science & Engineering

- **GPA:** 4.18/4.5, *Magna Cum Laude*

University of California, Berkeley (UC Berkeley) Jan 2025 – May 2025

Berkeley Global Access Program

Korea Advanced Institute of Science and Technology (KAIST) Aug 2024 – Dec 2024

Visiting Student

University of California, Berkeley June 2023 – Aug 2023

Summer Session Program

Research Experience

Neurophotronics Lab, GIST Gwangju, South Korea

Undergraduate Researcher (Advisor: Prof. Euiheon Chung) Sep 2025 – Present

- Proposed a cognitive framework that rule inference emerges as a hierarchical computation grounded in dynamical neural motifs. Demonstrated it in multi-task recurrent neural network (RNN) representations.
- Analyzed trial-to-trial shared variability in EEG and its alignment with the readout axis to account for age-related cognitive decline.
- Applied spectral graph theory on high-density multi-electrode array (HD-MEA) recordings and spiking neural networks (SNNs) to characterize mesoscale network dynamics.

The Computation and Language Lab, UC Berkeley Berkeley, United States

Research Assistant (Advisor: Prof. Steven Piantadosi) May 2025 – Aug 2025

- Provided a cognitive modeling framework using epsilon states from computational mechanics, showing their low-dimensional and causally relevant subspace in RNN and fMRI representations.

Brain Machine Intelligence Lab, KAIST Daejeon, South Korea

Undergraduate Researcher (Advisor: Prof. Sang Wan Lee) Sep 2024 – Dec 2024

- Supported study design, code review, and literature search for a project decoding prediction errors from EEG during model-based and model-free decision making.

BioComputing Lab, GIST Gwangju, South Korea

Undergraduate Researcher (Advisor: Prof. Sung Chan Jun) June 2024 – Aug 2024

- Designed and conducted a human cognitive task to investigate how label manipulation alters EEG neural correlates of deepfake discrimination.

Neurophotronics Lab, GIST Gwangju, South Korea

Undergraduate Researcher (Advisor: Prof. Euiheon Chung) Dec 2022 – Feb 2024

- Contributed to experimental setup and project conceptualization for HD-MEA neuronal network analysis.

Publications

Lee, H., Ashiquzzaman, A., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E. (in preparation). Spectral Graph Theoretical Characterization of Cortical Neuronal Networks using HD-MEA.

Lee, H., & Piantadosi, S. T. (manuscript in progress). Modeling Internal Representations of Task-Trained RNNs as Cognitive-State Automata.

Ashiquzzaman, A., **Lee H.**, Lee, E., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E. (submitted). Cortical Neuronal Network Developmental-State Transitions Detection and Characterization with High-Density Multi-Electrode Array.

Conferences

Lee, H., & Chung, E. (2026, Nov 14-18). *Motif-based hierarchical rule inference in recurrent neural networks* [Poster]. Neuroscience (Society for Neuroscience), Washington D.C., United States.

Lee, H., & Chung, E. (2026, May 7-9). *Influence of Hierarchical Task Inference on Representational Dynamics in Healthy and Aging-like RNNs under Hidden-Rule Conditions* [Poster]. The Korean Society of Medical & Biological Engineering Spring Annual Meeting, Jeju, South Korea.

Lee, H., & Chung, E. (2025, Nov 8). *Age-Related Reorganization of Shared Neural Variability and Readout Alignment During Cognitive Control* [Poster]. The 19th International Symposium for Aging, Gwangju, South Korea.

Lee, H., Ashiquzzaman, A., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E. (2025, Nov 6-8). *Spectral Graph Theoretical Characterization of Cortical Neuronal Networks using HD-MEA* [Poster]. The Korean Society of Medical & Biological Engineering Fall Annual Meeting, Gimhae, South Korea.

Teaching Experience

Computer Programming: Teaching Assistant	Mar 2026 – Present
Open GIST Science Lab: Programming Teaching Assistant	July 2026
Single Variable Calculus and Applications: Teaching Assistant	Mar 2026 – June 2026
PIUM: Science Tutoring Volunteer for the Suburbs	Sep 2022 – Nov 2022

Extracurricular Activities

Paper to Peer: Mathematics Paper Reading Club	Aug 2026 – Present
Gravity: Bio-imaging Mock Startup Project	June 2026 – Present
AlleoDreamy: AI Newsletter Publishing Team	Nov 2024 – Aug 2025
Google Developer Student Clubs: External Relations Leader	Sep 2023 – Feb 2024
Buddy Program: Volunteer for International Freshmen	Sep 2023 – Dec 2023
Korean I-Corps: AI Clothing Customization Mock Startup	Jun 2022 – Feb 2023
Asian Science Camp: Representative of Korea	June 2022

Skills

Experiment: Primary Neuron Culture, HD-MEA, Human Behavioral Tasks, EEG, Immunofluorescence, Calcium Imaging, fMRI analysis, Vibratome
Tools: ImageJ (Fiji), Git, Brainwave (3Brain), EEGLAB, L ^A T _E X, UCINET
Programming: Python, MATLAB, C, R, Java
Language: Korean (Native), English (Fluent; TOEFL 113/120)

Awards & Honors

National Scholarship for Science and Technology	Mar 2022 – Dec 2026
Trainee Professional Development Award (Travel Grant) <i>Society for Neuroscience</i>	Aug 2026
GIST-Fellowship for Innovative Research in Science & Technology	June 2026 - Aug 2026
Best Poster Award <i>The Korean Society of Medical & Biological Engineering Spring Annual Meeting</i>	May 2026
Best Poster Award <i>The 19th International Symposium for Aging</i>	Nov 2025
Study Abroad Program Scholarship	Jan 2025 – Mar 2025
Academic Excellence Scholarship	Sep 2024 – Dec 2024
AWS AI Award (1st place/200+ teams nationwide) <i>AI4GOOD Hackathon</i>	Mar 2024
Academic Excellence Scholarship	Mar 2022 – Dec 2023